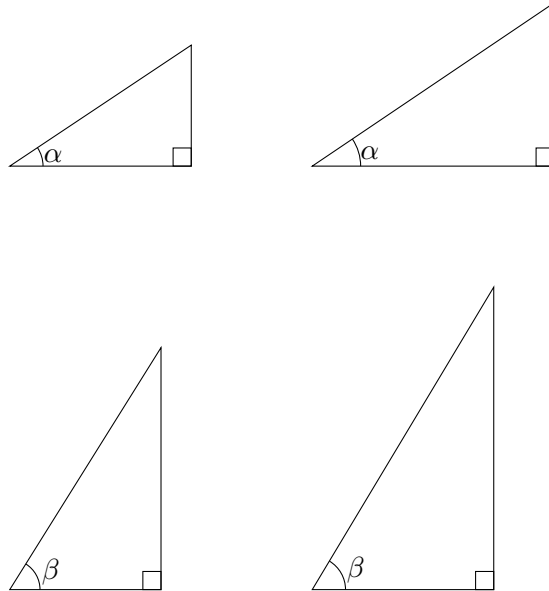


Exercise 1

For each triangle, mark the hypotenuse with H , the side opposite the marked angle with O , and the adjacent side with A . Calculate $\sin$ and $\cos$ of the marked angles. What do you observe regarding similar triangles?

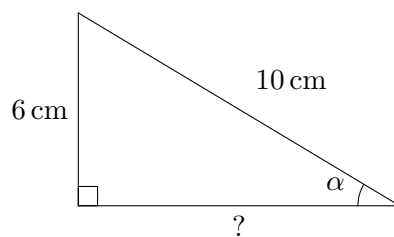


Exercise 2

A right triangle has a hypotenuse of 10 cm. The side opposite the angle α has length 6 cm.

1. Use Pythagoras' theorem to calculate the adjacent side.
2. Calculate $\sin(\alpha)$ and $\cos(\alpha)$.
3. Calculate $\sin^2(\alpha) + \cos^2(\alpha)$.

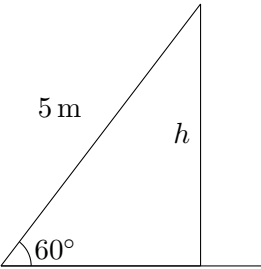
Do you think the result is just a coincidence?



Exercise 3

A ladder of length 5 m leans against a wall. The angle between the ladder and the ground is 60° .

How can you calculate the height h ? Write down an equation for h .



Exercise 4

Complete the table. Remember that $360^\circ = 2\pi$.

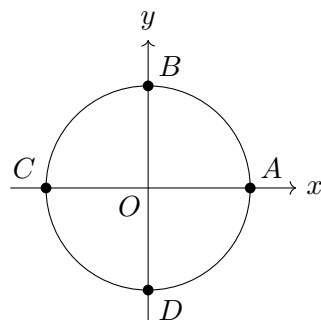
Degrees	Radians	Degrees	Radians
0°		180°	
30°		270°	
45°		360°	
60°			$\frac{3\pi}{2}$
90°			$\frac{\pi}{4}$

Exercise 5

The radius of the unit circle is 1. For an angle α , the point on the circle has coordinates

$$(\cos \alpha, \sin \alpha).$$

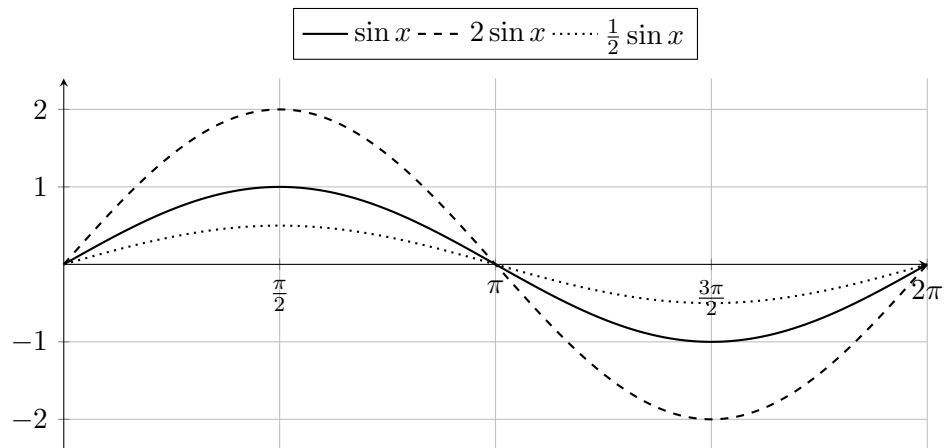
Label the points A – D with their coordinates and complete the table.



Angle	$\cos \alpha$	$\sin \alpha$
0		
$\frac{\pi}{2}$		
π		
$\frac{3\pi}{2}$		
2π		

Exercise 13: What does the factor in front do?

Compare the three graphs.



a) Complete the table.

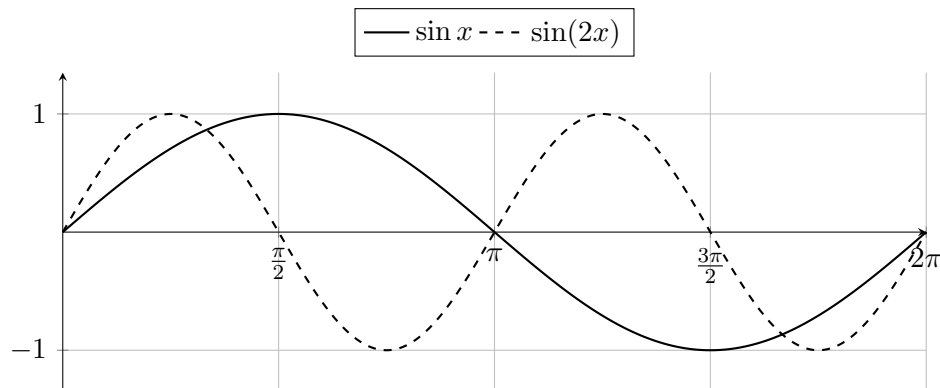
Function	Amplitude	Period
$\sin x$		
$2 \sin x$		
$\frac{1}{2} \sin x$		

b) What changes when the factor in front of the sine changes?

Part D continued: Frequency and Phase Shift

Exercise 14: What does the factor inside do?

Compare $y = \sin(x)$ and $y = \sin(2x)$.

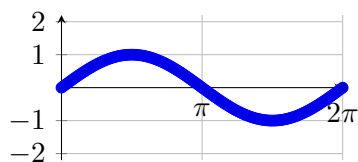


- How many complete oscillations does each graph make between 0 and 2π ?
- Which graph has the shorter period? _____
- Complete: The period of $\sin(2x)$ is _____
- Does the amplitude change? _____

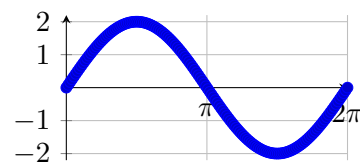
Exercise 15: Match equations and graphs

Match each graph A–D to one equation. Write the correct letter next to each equation.

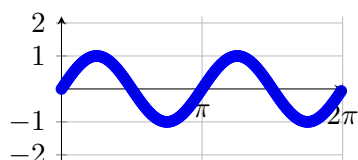
Graph A



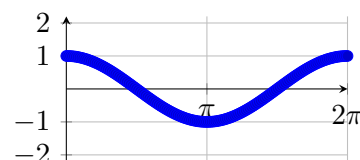
Graph B



Graph C



Graph D

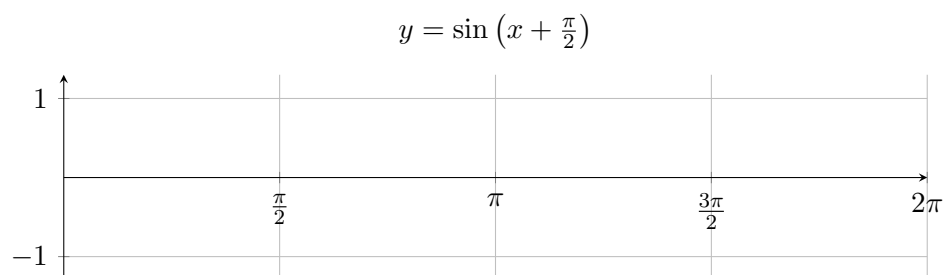
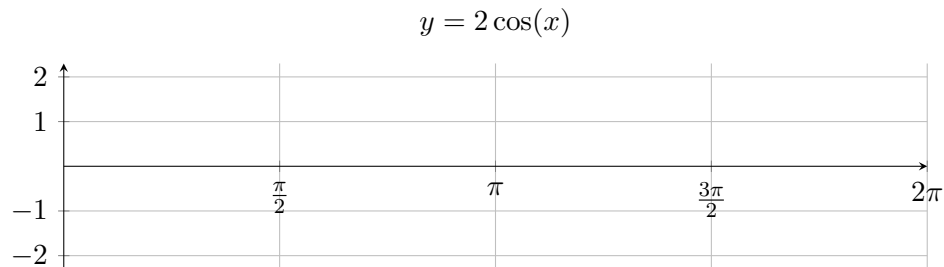


- | | | | |
|---|-------|---|-------|
| ☐ $y = \sin(x)$ | _____ | ☐ $y = \sin(2x)$ | _____ |
| ☐ $y = 2\sin(x)$ | _____ | ☐ $y = \cos(x)$ | _____ |

Part E: Sketching and Explaining

Exercise 16: Sketch the graphs

Sketch one complete period of each function. Mark the zeros, maxima, and minima.



Exercise 17: True or false?

Tick the correct box. Correct every false statement.

Statement	True	False
The sine function has period 2π .	☐	☐
The cosine function starts at 0.	☐	☐
The maximum value of $3 \sin(x)$ is 3.	☐	☐
The function $\sin(2x)$ has amplitude 2.	☐	☐
Sine and cosine have the same shape but are shifted horizontally.	☐	☐

Corrections:

Challenge: Describe a transformed sine function

Consider

$$f(x) = 3 \sin\left(\frac{x}{2} - \pi\right) + 1.$$

Without drawing the graph, determine:

Amplitude: _____

Period: _____

Horizontal

shift: Vertical shift: _____

Explain in one or two sentences how the graph differs from $y = \sin(x)$.

Remember: sine and cosine are not only triangle ratios. Their graphs are the basic mathematical shapes of oscillations and waves.

Exercise 6

Determine the period of the following functions and arrange them from the shortest period to the longest period.

a) $y = \sin(x) + 4$

b) $y = \sin\left(2x + \frac{\pi}{2}\right) + 1$

c) $y = \sin(3x)$

d) $y = \sin\left(\frac{3}{4}x\right)$

e) $y = \cos(4x - 1) - 2$

f) $y = \cos\left(\frac{1}{4}x\right)$

Exercise 7

Describe what happens in the following cases:

- The graph changes from $y = \cos(x)$ to $y = \cos(2x) + 2$.
- The graph changes from $y = \cos(x)$ to $y = \cos(0.5x) - 3$.
- The graph changes from $y = \sin(x)$ to $y = \sin(x - \pi)$.
- The graph changes from $y = \cos(x)$ to $y = \cos\left(x + \frac{\pi}{2}\right)$.

Exercise 8

Choose the correct function.

1. The graph has amplitude 1 and period π :

☐ $\sin(x)$

☐ $\sin(2x)$

☐ $\sin(0.5x)$

☐ $\sin(3x)$

2. The graph has amplitude 1 and period 4π :

☐ $\cos(x)$

☐ $\cos(2x)$

☐ $\cos(0.5x)$

☐ $\cos(3x)$

3. Which function is shifted to the right by π compared to $\sin(x)$?

☐ $\sin(x + \pi)$

☐ $\sin(2x)$

☐ $\sin(x - \pi)$

☐ $\cos(x)$

4. Which function has the shortest period?

☐ $\sin(x)$

☐ $\sin(5x)$

☐ $\sin(2x)$

☐ $\sin(0.25x)$

Exercise 9

Complete the functions — fill in the missing numbers.

1. The period should become π :

$$y = \sin(\text{_____}x)$$

2. The period should become 4π :

$$y = \sin(\text{_____}x)$$

3. The period should become $\frac{\pi}{2}$:

$$y = \sin(\text{_____}x)$$

4. The graph should be shifted to the right by π :

$$y = \sin(x \pm \text{_____})$$

5. The graph should be shifted to the left by $\frac{\pi}{2}$:

$$y = \cos(x \pm \text{_____})$$

Modelling Real Situations

1. Two swinging pendulums

Two pendulums swing with the same amplitude (1 cm).

- Pendulum A completes one full swing every 4 seconds.
- Pendulum B completes one full swing every 2 seconds.
- Both start at their highest position at $t = 0$.

- a) Write down a cosine function for each pendulum.
- b) Which pendulum has the longer period?
- c) Which pendulum moves faster?
- d) Which graph is stretched more along the time axis?

Hint: It might be helpful to sketch the graphs.

2. Build your own oscillation

Your oscillation should have:

- amplitude 1,
- period π ,
- shifted by $\frac{\pi}{2}$ to the right.

Write a possible sine function describing this wave.

Is there another possibility to express the function, e.g. as a cosine function?
